

Advances in the drug treatment of Alzheimer's disease: pathophysiology and mechanisms of action

Joseph Nowell,¹ Harry Crook,¹ Mony J de Leon,² Paul Edison¹



¹Department of Brain Sciences, Faculty of Medicine, Imperial College London, Hammersmith Campus, London, UK

²Brain Health Imaging Institute, Department of Radiology, Weill Cornell Medicine, New York, NY, USA

Correspondence to: P Edison
paul.edison@imperial.ac.uk

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ABSTRACT

Alzheimer's disease, the leading cause of dementia, is a multifactorial disorder involving amyloid beta (A β) and tau deposition, impaired glucose metabolism, neuroinflammation, mitochondrial dysfunction, insulin resistance, and progressive brain atrophy. Anti-A β therapies have shown clinical efficacy and are licensed in several countries. Amyloid related imaging abnormalities remain a key safety concern, and reimbursement varies across healthcare systems. The mechanisms underlying continued cognitive decline after A β clearance remain unclear and might be independent of amyloid pathology. Because Alzheimer's disease involves multiple pathological processes, effective management will likely require combination treatments addressing tau aggregation, neuroinflammation, synaptic loss, and metabolic dysfunction. Numerous compounds targeting these mechanisms are currently in late stage development for both early and advanced disease. These emerging approaches represent a shift toward multimodal, disease modifying strategies designed to improve patient outcomes and quality of life. Here, we review recent therapeutic advances in Alzheimer's disease and provide perspectives on novel treatment strategies.

Introduction

Alzheimer's disease is a progressive neurodegenerative disorder affecting millions worldwide, causing major societal and economic burden.¹ Neuropathogenic changes appear 15-20 years before onset, followed by a prodromal mild cognitive impairment phase and subsequent progression to Alzheimer's disease.² Patients with amnesic mild cognitive impairment (subjects with impaired episodic memory) show early memory deficits progressing to widespread cognitive, behavioral, and functional decline.³

Alzheimer's disease involves not only amyloid β (A β) plaques and tau tangles but also synaptic dysfunction, neuroinflammation, oxidative stress, and metabolic impairment. Until recently, treatment focused solely on symptomatic relief by using cholinesterase inhibitor drugs and N-methyl-D-aspartate receptor antagonist drugs. After decades of research, the approval of the anti-A β drugs lecanemab and donanemab marks the first disease modifying options for Alzheimer's disease. This review explores current and emerging therapeutic strategies targeting A β , tau, neuroinflammation, neuromodulation, and metabolism, and outlines how new diagnostic criteria integrate into anti-A β treatment pathways.

Epidemiology

Alzheimer's disease, the most common form of dementia accounting for 60% to 80% of cases,⁴

affects over 55 million people worldwide and is the seventh leading cause of death.^{5,6} With global aging, cases are projected to triple by 2050, posing a major healthcare burden.⁷ Fewer than 1% of cases stem from autosomal dominant mutations in APP (amyloid precursor protein), PSEN1 (presenilin 1), or PSEN2, while most are sporadic, with age and the APOE ϵ 4 allele as key risk factors, with women facing over twice the risk of men.^{4,8} The Lancet Commission has identified 14 modifiable risk factors for dementia, including less education, hypertension, hearing impairment, smoking, obesity, depression, physical inactivity, diabetes, low social contact, excessive alcohol consumption, traumatic brain injury, air pollution, untreated vision loss, and high levels of low density lipoprotein cholesterol.⁹

Diagnosis remains difficult, leaving many undetected cases, especially in underrepresented populations, with complex interactions between age, sex, genetics, and environment shaping its epidemiology.

Sources and selection criteria

Firstly, we conducted a search of the ClinicalTrials.gov database with the following criteria: Alzheimer's disease, interventional study types, with results, phase 2, 3, or 4 studies that assessed the efficacy of intervention, first posted from 1 January 2010. To identify the most recent drug substances in

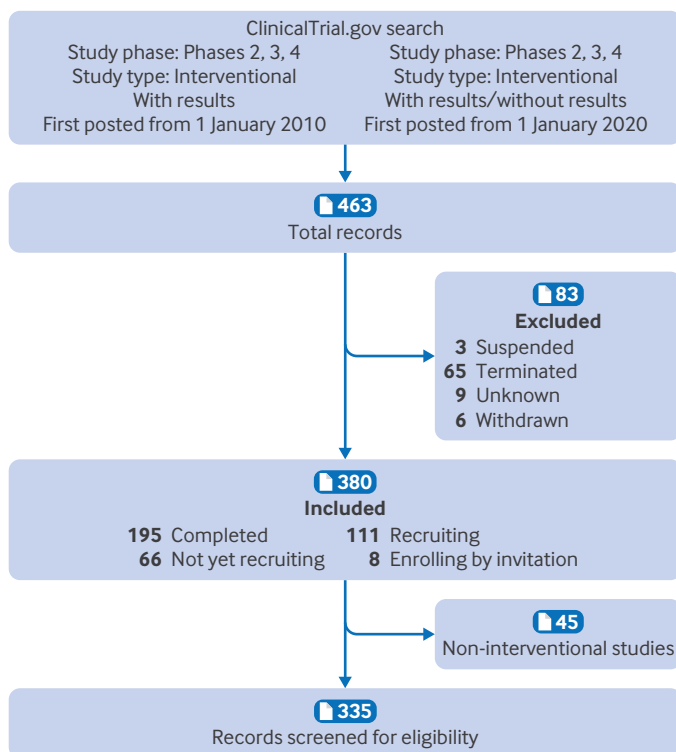


Fig 1 | Selection criteria of the trials from ClinicalTrials.gov

development, we then conducted a search of the ClinicalTrials.gov database with the following criteria: Alzheimer's disease, interventional study types, with results and without results, phase 2, 3, or 4 studies that assessed the efficacy of intervention, first posted from 1 January 2020 (fig 1).

Next, we searched all eligible records of the novel interventions found by using the ClinicalTrials.gov database search. For example, we performed a MEDLINE (PubMed) database search using the search terms "[intervention/compound/drug]" and "Alzheimer's disease." We screened abstracts and then entire papers for eligibility based on whether they were intervention studies and registered on clinical trials.gov, and then manually searched through the reference lists of the included papers to build a comprehensive library of information. All interventions were manually screened with the five most common disease modifying categories identified as: "anti-amyloid," "anti-tau," "neuroinflammation," "neuromodulation," and "metabolic."

Alzheimer's disease pathophysiology

Alzheimer's disease is defined by A β plaque and neurofibrillary tau tangle formation. According to the amyloid cascade hypothesis, A β —produced by β secretase cleavage of APP into A β_{40} and the more pathogenic A β_{42} —initiates tau pathology and neurodegeneration.^{14,10} The interaction between A β and tau is considered central to progression of the disease.¹¹ The probabilistic model refines this view, suggesting that A β 's pathogenicity depends on genetic and environmental factors,¹² with

deterministic progression in familial cases and variable outcomes in sporadic forms.

Beyond A β and tau, Alzheimer's disease involves neuroinflammation, synaptic dysfunction, metabolic deficits, and insulin resistance (fig 2). Activated glial cells lose homeostatic control, releasing inflammatory mediators and damaging synapses.¹³ Synaptic degeneration appears early and drives cognitive and behavioral decline. Metabolic dysfunction and impaired insulin signaling are also early features of Alzheimer's disease.¹⁴ These abnormalities worsen cognitive impairment and promote A β and tau pathology progression.¹⁵

The complexity of Alzheimer's disease has made developing effective treatments to modify the disease challenging.¹⁶ Research is focusing on multi-targeted approaches, either by using single drug substances that affect multiple pathways or combinations of drug substances that target individual pathological features. Here, we discuss different drugs for the treatment of Alzheimer's disease and their progress in clinical trials.

Diagnosis

Biomarkers

New diagnostic criteria incorporate core biomarkers of Alzheimer's disease neuropathologic change and non-specific biomarkers important in the disease pathogenesis but also occur in other neurodegenerative conditions.¹⁷ Core 1 biomarkers include A β positron emission tomography (PET) and cerebrospinal fluid (CSF)/plasma markers that become abnormal simultaneously, including phosphorylated and secreted tau (especially phosphorylated tau 217), which can help establish diagnosis. Core 2 biomarkers include tau-PET and other CSF/plasma markers of tau (phosphorylated tau 205) that provide prognostic information and improve diagnostic confidence. Non-specific biomarkers include measures of neural injury, dysfunction, or degeneration of neuropil (eg, [¹⁸F] fluorodeoxyglucose-PET and structural magnetic resonance imaging (MRI)) and inflammation (glial fibrillary acidic protein) (table 1).

Alzheimer's disease as a clinical-biological construct

In contrast to the National Institute on Aging and Alzheimer's Association research framework, which defines Alzheimer's disease as a biological construct based on A β and tau biomarkers, the International Working Group advocates for defining Alzheimer's disease as a clinical-biological construct.¹⁹ Although it acknowledges the importance of biomarkers, this perspective emphasizes the necessity of clinical manifestation for a diagnosis of Alzheimer's disease. The International Working Group defines Alzheimer's disease as a clinical-biological entity requiring both the presence of positive pathophysiological biomarkers (A β and tau) and a specific clinical phenotype consistent with Alzheimer's disease that commonly include amnesic syndrome of the

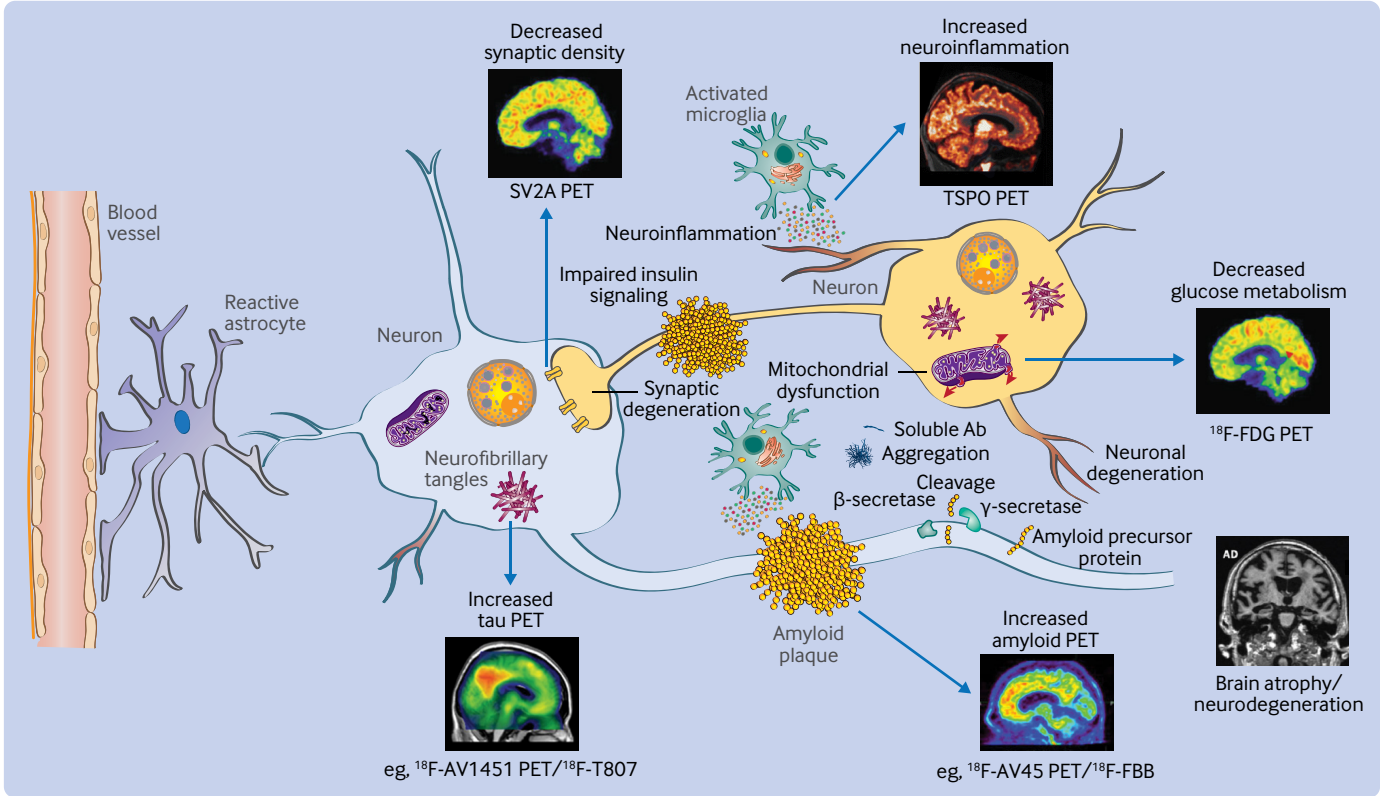


Fig 2 | Multifaceted nature of Alzheimer's disease. Extracellular amyloid plaques aggregate after the cleavage of amyloid precursor protein to form A β monomers that convert to higher order oligomers, protofibrils, and fibrils. Intracellular tau tangles form and influence neuronal degeneration. Several other pathological processes are central features of Alzheimer's disease, including synaptic degeneration, neuroinflammation, and mitochondrial dysfunction

hippocampal type, logopenic aphasia, and posterior cortical atrophy or, less commonly, corticobasal syndrome and behavioral and dysexecutive variants. The International Working Group approach enables the identification of two categories for cognitively normal individuals with positive biomarkers: cognitively normal A β positive at risk of developing Alzheimer's disease; and cognitively normal with specific Alzheimer's disease pattern of biomarkers with an almost deterministic and very high risk of Alzheimer's disease.

Radiotracers

Positron emission tomography imaging utilizes radiotracers to measure metabolic processes within the body including the abnormal protein aggregation, synaptic dysfunction, and neuroinflammation. A β -PET tracers allow direct evaluation of the total load and spatial distribution of A β protein fibrillary deposits.²⁰ Generally considered the gold standard of A β -PET, [¹¹C]-labeled Pittsburgh compound B displays a relatively high affinity for all forms of A β .²¹ Three [¹⁸F]-labeled A β radiotracers have been

Table 1 Criteria for diagnosis and staging Alzheimer's disease ¹⁸				
Biomarker category	Process	CSF/plasma analysis	Imaging	Usage
Aβ plaques	Aβ proteinopathy	Aβ ₄₂	Amyloid-PET	Establish diagnosis
Soluble tau	Phosphorylated and secreted Alzheimer's disease tau	p-tau 217, p-tau 181, p-tau 231		Inform clinical decision making
Biomarkers of Alzheimer's pathology	Alzheimer's disease tau proteinopathy	Microtubule binding region tau 243, other phosphorylated tau forms (eg, p-tau 205), non-phosphorylated mid-region tau fragments	tau-PET	Prognostic information Confirm diagnosis/stage biological disease severity
Biomarkers of non-specific processes involved in Alzheimer's disease pathophysiology	Neurodegeneration	Neurofilament light	Anatomic MRI, fluorodeoxyglucose PET	Clinical use when considering the cause of symptoms, prognosis, and treatment decisions Exclusion criteria for clinical trials
	Inflammation	Glial fibrillary acidic protein		
Biomarkers of non-Alzheimer's disease copathology	α-synuclein	α-synuclein seed amplification assay		Role in clinical care
	Vascular brain injury	Infarction on MRI or CT, white matter hyperintensity		
MRI=magnetic resonance imaging; CT=computed tomography				

MRI=magnetic resonance imaging; CT=computed tomography

approved for clinical use: [^{18}F]florbetapir, [^{18}F]flutemetamol, and [^{18}F]florbetaben.

The first fluid based biomarkers used in Alzheimer's disease clinical trials were CSF biomarkers owing to their proximity to the brain and relatively high concentration of markers related to Alzheimer's disease.²² Alterations in CSF $\text{A}\beta_{42}/\text{A}\beta_{40}$ ratios characterize initial stages of Alzheimer's disease, with $\text{A}\beta_{42}/\text{A}\beta_{40}$ ratios effectively detected in plasma samples. Accounting for $\text{A}\beta_{40}$ ratio improves the sensitivity by controlling for individual differences in the production of $\text{A}\beta$.²² CSF $\text{A}\beta_{42}/\text{A}\beta_{40}$ effectively distinguishes patients with Alzheimer's disease from patients with other dementia related disorders and can accurately predict conversion of mild cognitive impairment to Alzheimer's disease.²³ In vivo measures of tau include tau PET and CSF/plasma markers.

[^{18}F]Flortaucipir is the only Food and Drug Administration and European Medicines Agency approved tau-PET radiotracer for clinical use and mirrors the stereotypical Braak staging pattern.²⁴ [^{18}F]-PI-2620 PET is a second generation tau-PET tracer with high binding affinity to pathological tau depositions and low off-target binding and has received FDA Fast Track designation for Alzheimer's disease.²⁵ CSF markers of phosphorylated tau (eg, p-tau 217) are the most specific for Alzheimer's disease, detected at low concentrations in plasma.²⁶ Changes in CSF total tau and p-tau occur before $\text{A}\beta$ -PET positivity.²⁷ A novel CSF marker of insoluble tau aggregates, microtubule binding region of tau containing the residue 243 (microtubule binding region tau 243), shows strong associations with tau-PET and a higher correlation with cognition than other established CSF tau measures.²⁸

Recent efforts on developing cost effective and accessible blood based biomarkers have shown high accuracy of p-tau 181 and p-tau 217 to detect Alzheimer's disease, comparable to CSF counterparts.²⁹ Plasma p-tau 181, p-tau 217, and p-tau 231 typically increase in the early stages of Alzheimer's disease as $\text{A}\beta$ /tau aggregate in the brain, although at more advanced stages, concentrations of these soluble forms plateau and even decline.^{30 31} p-tau 217 shows higher diagnostic accuracy than p-tau 181, and can detect $\text{A}\beta$, tau, and cognitive abnormalities.³²

Current therapies

Anti- $\text{A}\beta$ therapy

The amyloid cascade hypothesis, eccentric to research on Alzheimer's disease,^{10 33-35} proposes that $\text{A}\beta$ is generated from APP cleavage by BACE1 (β -secretase) and γ -secretase, producing mainly $\text{A}\beta_{40}$ and the aggregation prone $\text{A}\beta_{42}$.³⁶ $\text{A}\beta$ monomers form plaques by aggregating into oligomers, protofibrils, and fibrils.³⁷ Therapeutic strategies target various stages of $\text{A}\beta$ formation, from BACE1 and γ -secretase inhibition to directly targeting $\text{A}\beta$ conformations (fig 3).

Between 2004 and 2021, 23 unique anti- $\text{A}\beta$ drug substances were tested and failed to meet the primary

endpoint in phase 2 or phase 3 clinical evaluation.³⁸ Many anti- $\text{A}\beta$ trial failures might reflect compound non-specificity, $\text{A}\beta$'s physiological roles, timing of intervention, and uncertainty over which $\text{A}\beta$ forms are neurotoxic.

Soluble nonfibrillar $\text{A}\beta$ oligomers are considered the primary neurotoxic species in early Alzheimer's disease, correlating with symptom severity.³⁹⁻⁴¹ In vitro and in vivo studies show that $\text{A}\beta$ oligomers impair synaptic transmission, inhibit long term potentiation, enhance long term depression, and reduce dendritic spine density. These effects, observed in animal models, are largely attributed to $\text{A}\beta$ dimers disrupting cell membrane integrity.⁴² By contrast, fibrillar $\text{A}\beta$ in plaques also causes neuronal and axonal loss.⁴³ Overall, disease severity aligns more closely with soluble $\text{A}\beta$ levels than with aggregated plaques.⁴⁰

Targeting specific $\text{A}\beta$ conformations within the optimal therapeutic window is crucial, because the success of lecanemab and donanemab reinforce $\text{A}\beta$'s central role in Alzheimer's disease, providing impetus for future disease modifying therapies.⁴⁴ Donanemab binds the pyroglutamate modified $\text{A}\beta$ present in plaques, facilitating their clearance, whereas lecanemab primarily targets soluble protofibrils. These pharmacodynamic differences explain their distinct treatment protocols—donanemab is discontinued once $\text{A}\beta$ PET imaging becomes negative, while lecanemab is continued beyond 18 months to sustain anti-oligomeric activity and prevent reaccumulation of toxic soluble $\text{A}\beta$ species.

Mechanism of targeting amyloid in Alzheimer's disease trajectory

Passive immunotherapy

Recent anti- $\text{A}\beta$ therapy efforts have focused on passive immunotherapy using intravenously administered monoclonal antibodies that target specific $\text{A}\beta$ conformations. Despite encouraging results from some anti- $\text{A}\beta$ therapies, several phase 3 trials, including Graduate 1 and 2 with gantenerumab, failed to show meaningful cognitive improvement despite substantial $\text{A}\beta$ reduction.⁴⁵ Likewise, solanezumab, crenezumab, and bapineuzumab produced negative outcomes, highlighting that $\text{A}\beta$ clearance alone might be insufficient to achieve noticeable clinical benefit in Alzheimer's disease.⁴⁶⁻⁴⁸

After many unsuccessful trials, the FDA approved lecanemab (2023) and donanemab (2024).⁴⁹ Aducanumab (2021) received accelerated approval but was later discontinued. In November 2024, the EMA recommended lecanemab for early Alzheimer's disease in individuals with one or no copy of the APOE $\epsilon 4$ allele.⁵⁰ In July 2025, donanemab was recommended for the same indication.⁵¹

Trial evidence

Two randomized phase 3 trials (Engage, n=1638; and Emerge, n=1647) evaluated the efficacy of aducanumab in early Alzheimer's disease.^{52 53}

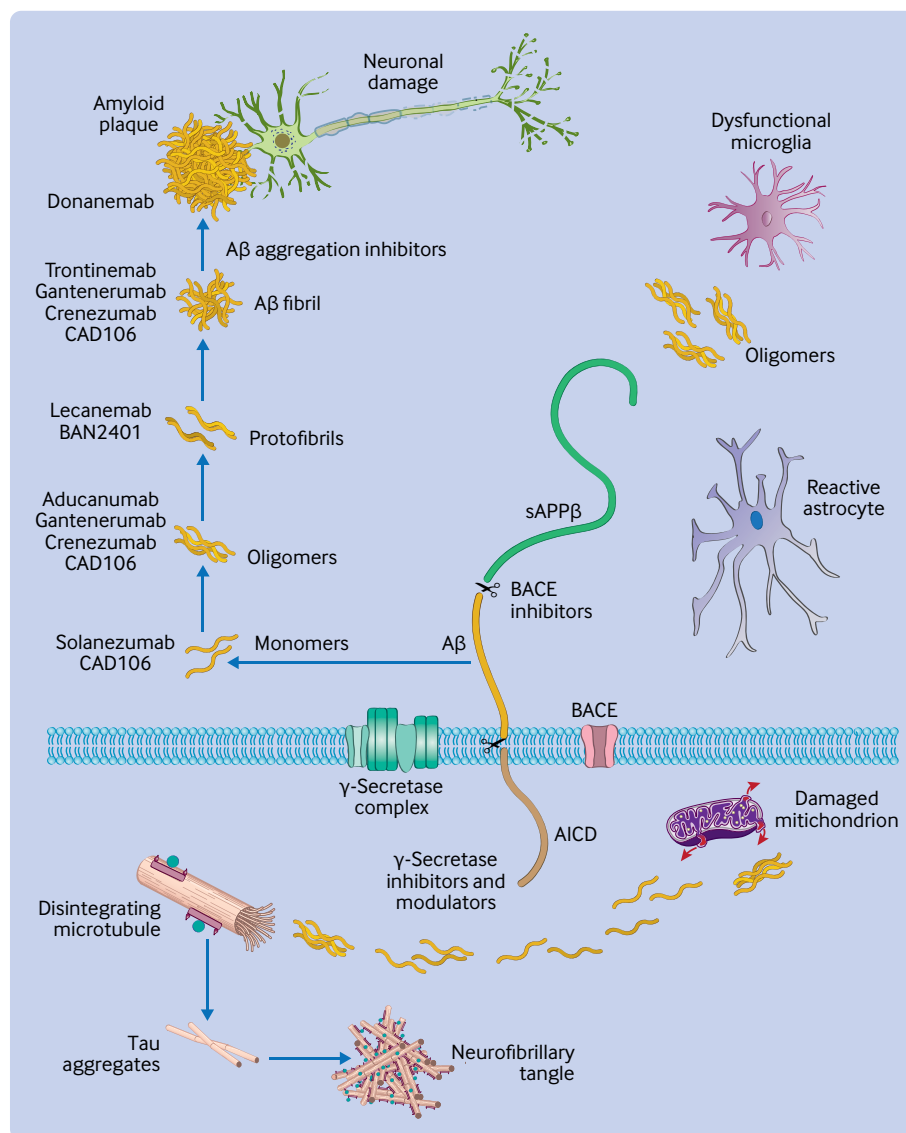


Fig 3 | Anti-A β therapy. Therapies under development include BACE1 inhibitors and γ -secretase modulators/inhibitors, as well as targeting A β monomers, oligomers, protofibrils, fibrils, and plaques. Other agents acting through indirect or non-immunologic pathways are described in the text but not illustrated here. The “oligomers” depicted on the right of the figure include both soluble A β oligomers and protofibrils, and that protofibrillar A β_{42} is the main inflammatory trigger relevant to lecanemab’s mechanism of action

Participants were allocated to receive a low (3 or 6 mg/kg), or high (10 mg/kg) dose of aducanumab or placebo once every 4 weeks for a period of 76 weeks. Based on the futility analysis of the pooled interim data, both trials were terminated early. However, analysis of each trial on the primary endpoint of the clinical dementia rating sum of boxes showed that there was a benefit of high dose aducanumab compared with placebo in Emerge (difference -0.39 points ν placebo; $P=0.012$) but not in Engage (-0.03 ; $P=0.83$). The most common adverse event was the incidence of amyloid-related imaging abnormalities with edema or effusions (ARIA-E) in 35% to 36% of high dose groups and 26% of low dose groups.

A double blind, phase 3 trial evaluated the effect of intravenous 10 mg/kg lecanemab every two

weeks.^{54 55} 1795 participants with early Alzheimer’s disease were randomized to receive lecanemab or placebo for 18 months. The primary endpoint showed a moderate reduction in the decline in clinical dementia rating sum of boxes score that favored lecanemab (difference -0.45 points ν placebo; $P<0.001$). Treatment was associated with adverse events: ARIA with hemosiderin deposits (ARIA-H) or ARIA-E occurred in 17.7% and 12.6% of participants, respectively. No deaths were considered related to lecanemab treatment or associated with ARIA. Serious adverse events occurred in 14.0% of participants receiving lecanemab compared with 11.3% in the placebo group. Reactions related to infusion were the most common serious adverse event (reported in 1.2% of the participants treated with

Box 1: Lecanemab prescribing recommendations⁵⁶**Inclusion criteria**

- Diagnosis of mild cognitive impairment or mild Alzheimer's disease dementia
- Amyloid positive (indicated by cerebrospinal fluid or amyloid positron emission tomography)
- Mini mental state examination score 22-30^{*}
- Stable medication[†] (cognitive enhancing agents)
- Care partner or family member supervision

Exclusion criteria

- Any condition that could be contributing to cognitive impairment
- More than four microhaemorrhages
- Evidence of non-Alzheimer's disease dementia
- Recent history (within 12 months) of stroke or transient ischemic attacks or seizures
- Any confounding psychiatric diagnosis or symptoms
- Major depression (Geriatric depression scale score >8 at screening)
- Any immunological disease not adequately controlled
- Bleeding disorder not adequately controlled
- Patients taking anticoagulants
- Unstable medical conditions that interfere with lecanemab therapy

Dose

- 10 mg/kg once every 2 weeks

Administration

- Infusion bag containing 250 mL of sodium chloride 0.9% solution for injection administered through an intravenous line with a terminal low-protein binding 0.2 micron in-line filter

Infusion time

- Approximately 1 hour

Monitoring

- Patient monitoring for 3 hours after the first infusion and next day phone call. Reduced to 2 hours for the second and third doses, and then 30 minutes thereafter if no reactions occur

ARIA risk

- APOE genotyping to discuss enhanced ARIA risk in APOE4 homozygotes

ARIA monitoring

- MRI scans after the 5th, 7th, 14th, and 26th infusions

^{*}Or an equivalent level of impairment as determined by other validated measures such as the Montreal cognitive assessment or clinical dementia rating sum of boxes.

[†]Indicates that a participant has been taking standard Alzheimer's drugs (such as cholinesterase inhibitors or memantine) on a consistent dose and schedule before starting lecanemab. Maintaining an unchanged regimen helps ensure that any cognitive or functional changes observed during therapy are related to lecanemab rather than recent medication adjustments.

lecanemab). Numerical improvements were shown in CSF markers of A β , tau, and neurogranin, but not neurofilament light, a measure of neuroaxonal damage.

International recommendations

Lecanemab is approved for the treatment of early Alzheimer's disease in the US, UK, Europe, South Korea, Japan, China, Hong Kong, and the UAE.⁵¹ The US appropriate usage criteria are outlined in box 1.⁵⁶ The recommendations serve as essential tools for translating clinical trial data into real world practice, but they are not universally applicable.⁵⁷ Their relevance depends on each healthcare system's diagnostic infrastructure, biomarker availability, and monitoring capabilities. The US recommendations have shaped early implementation,⁵⁶ yet guidelines specific to each region have since emerged in Europe and South Korea,^{58 59} which reflect the local medical norms and resource constraints.

The risk of intracerebral hemorrhage associated with lecanemab is heightened in patients taking anticoagulation drugs. Consequently, all major recommendations advise against concurrent use of vitamin K antagonists or direct oral anticoagulants. Low dose antiplatelet therapy is generally permissible, if patients are counselled and monitored closely. Where anticoagulation becomes unavoidable, therapy should be interrupted and re-evaluated.

A randomized phase 3 trial of 1795 people found that lecanemab reduced markers of amyloid in early Alzheimer's disease and resulted in moderately less decline on measures of cognition and function than placebo at 18 months. However, adverse effects were reported (infusion-related reactions in 26.4%, amyloid-related imaging abnormalities with edema or effusions in 12.6%).⁶⁰ If lecanemab is prescribed, ongoing benefit should be assessed after 18 months. Most guidance recommends structured reassessment

at this point, combining cognitive outcomes, caregiver input, and imaging safety data. Continued therapy might be reasonable where benefit persists and adverse events remain manageable, though longer term strategies—including lower frequency maintenance dosing—are still being defined.

Differences among international recommendations highlight the influence of healthcare context: US guidelines allow cautious use in APOE ϵ 4 homozygotes,⁵⁶ European guidance generally discourages it,⁵⁸ and Korean recommendations occupy a middle ground.⁵⁹ Anticoagulant use is universally contraindicated for full dose anticoagulation, with careful allowance for standard antiplatelet use. US and Korean frameworks require A β PET or CSF analysis. European protocols increasingly incorporate validated plasma biomarkers to expand access. The US model calls for imaging before selected infusions (5th, 7th, 14th), Europe favors earlier scans to detect ARIA sooner, and Korea follows a hybrid schedule. European guidance

typically includes an 18 month discontinuation point if clinical progression occurs, whereas US and Korean frameworks emphasize individualized reassessment and potential continuation.

Before initiating anti-A β therapy such as lecanemab, clinicians should verify A β pathology, perform APOE genotyping, and exclude patients requiring anticoagulation. In ϵ 4 homozygotes, treatment should be reserved for research or exceptional cases with informed consent and intensive monitoring. Therapy should be reassessed after 18 months to determine whether continuation remains justified. Ultimately, the recommendations must be adapted to local diagnostic capabilities and healthcare structures rather than applied as uniform global standards.

Leqembi Iqlik

Leqembi Iqlik is the first anti-A β therapy that allows patients to continue treatment at home by using subcutaneous injections after completing 18

Box 2: Donanemab prescribing recommendations⁶⁵

Inclusion criteria

- Diagnosis of mild cognitive impairment or mild dementia stage of disease
- Amyloid positive
- Mini mental state examination score 20–28^{*} (evaluated in Trailblazer-Alz 2 trial)
- Stable medication[†] (cognitive enhancing agents)
- Care partner or family member supervision

Exclusion criteria

- Any condition that could be contributing to cognitive impairment
- Evidence of non-Alzheimer's disease dementia
- Previous treatment with a passive anti-A β immunotherapy
- Poor venous access
- History of cancer within the past five years
- History of immunologic disease
- More than four cerebral microbleeds, at least one area of cortical superficial siderosis or a major vascular contribution
- Patients with a bleeding disorder that is not under adequate control
- Patients taking anticoagulants
- Any previously known serious hypersensitivity to anti-A β , PET, or MRI

Dose

- 700 mg for the initial three doses followed by 1400 mg doses thereafter, once every 4 weeks

Administration

- Intravenous infusion using a needle, dilution to a final concentration of 4 mg/mL to 10 mg/mL with sodium chloride 0.9% solution for injection

Infusion time

- Approximately 30 minutes

Monitoring

- Patient monitoring for a minimum of 30 minutes

ARIA risk

- APOE genotyping to discuss enhanced ARIA risk in APOE4 homozygotes

ARIA monitoring

- MRI scans before initiating treatment and after the 2nd, 3rd, 4th, and 7th infusions, and before the 12th dose

^{*}Or an equivalent level of impairment as determined by other validated measures such as the Montreal cognitive assessment or clinical dementia rating sum of boxes.

[†]Indicates that a participant has been taking standard Alzheimer's drugs (such as cholinesterase inhibitors or memantine) on a consistent dose and schedule before starting donanemab. Maintaining an unchanged regimen helps ensure that any cognitive or functional changes observed during therapy are related to donanemab rather than recent medication adjustments.

months of intravenous lecanemab. After the initial intravenous phase (10 mg/kg every two weeks), patients can either switch to monthly intravenous infusions or weekly 360 mg subcutaneous doses using the Iqlik autoinjector. In the Clarity Alzheimer's disease open label extension, subcutaneous dosing achieved approximately 14% greater A β reduction and 11% higher drug exposure than intravenous treatment, remaining within bioequivalence limits. Injection reactions were mild, and ARIA rates were comparable between intravenous and subcutaneous injections, though slightly higher numerical rates were noted in the smaller subcutaneous dosing group; these differences were descriptive and not statistically tested.⁶¹

Donanemab

Donanemab for early symptomatic Alzheimer's disease showed a benefit in a phase 3 investigation.⁶² The primary endpoint indicated that donanemab slowed clinical progression on the integrated Alzheimer's disease rating scale score compared with placebo. Donanemab noticeably slowed clinical progression over 76 weeks in individuals with low or medium tau pathology, as well as in the overall population (including those with high tau pathology); however, treatment did not noticeably reduce frontal tau burden as measured by tau PET imaging. ARIA was present in 24% of those treated with donanemab. In the donanemab group, three participants with serious ARIA subsequently died. Data from a phase 3 investigation showed that donanemab is superior to aducanumab in the clearance of A β at six months.⁶³ Based on the promising phase 3 results, donanemab is approved in numerous countries (eg, USA,⁶⁴ China, and Japan). Box 2 outlines the recommendations for prescribing donanemab.^{65 66} In interim phase 1 trials involving 41 participants, treatment with 2800 mg remternetug (an enhanced analogous of donanemab) monthly showed rapid reductions of A β plaques.⁶⁷

Amyloid related imaging abnormalities

ARIA was first defined by the Alzheimer's Association Research Roundtable Workgroup in 2011, after a phase 1 trial of bapineuzumab in 2010.⁶⁸ ARIA encompasses two distinct forms: ARIA-E (edema) and ARIA-H (microhemorrhages).

ARIA-E is characterized by vasogenic edema, identifiable through a transient increase in signal intensity on fluid attenuated inversion recovery MRI images.⁶⁹ This edema manifests in the leptomeningeal/sulcal space, where the hyperintensity is attributed to fluid effusion or extravasation from surrounding blood vessels, and parenchymal changes indicative of edema.

ARIA-H is characterized by hemosiderin deposition resulting from the leakage of blood products, leading to small superficial siderosis and microhemorrhages detectable on T2-weighted MRI sequences or susceptibility weighted imaging.⁶⁹ Microhemorrhages appear as small focal round hypointensities (usually <10 mm) within the

brain parenchyma. Superficial siderosis is seen as curvilinear hypointensities on the brain's surface within the meningeal space.

About 74% of ARIA events are asymptomatic, while 6.1% to 39.3% present with the most common neurological symptoms, such as headaches, that usually resolve after dose adjustment.^{69 70} Most symptoms are transient and diminish by stopping treatment or reducing the dose. Severe ARIA events are uncommon, occurring in approximately 0.5% to 3% of treated patients, and may include clinically significant brain edema, seizures, or, rarely, death (reported in <0.5% of cases). APOE ϵ 4 carriers, older patients, and patients with microhemorrhages, taking antithrombotic drugs, or previous strokes have a higher risk of severe adverse events.⁷¹

Emerging treatments

Active immunotherapy

Active immunotherapy, or vaccination, works by introducing A β or its fragments to trigger an immune response.⁷² Antigen presenting cells process the antigen, activating CD4+ T cells, which then stimulate B cells to produce A β -specific antibodies.⁷³ Compared with passive therapies, active immunization requires fewer doses and promotes long term endogenous antibody production.⁷² It is considered a potentially more cost effective approach to targeting A β .⁷⁴ However, early clinical trials, such as AN1792, were halted after 6% of participants developed meningoencephalitis.

Several active immunotherapy candidates show improved safety but mixed efficacy in slowing cognitive decline and reducing A β plaque burden in phase 1/2 trials.⁷⁵ The ACI-24 vaccine showed beneficial effects on A β plaque load and microglial phenotypes in preclinical studies and was well tolerated in a randomized trial of young adults with Down's syndrome.^{74 76} Additionally, the first human phase 1/2a study of autologous A β mutant peptide-pulsed dendritic cells (ALZN002) is ongoing in participants with mild to moderate Alzheimer's disease (NCT05834296).

BACE1 inhibition

BACE1 inhibitor drugs block β -secretase cleavage of APP, reducing A β ₄₀ and A β ₄₂ production by binding to BACE1's active site.^{77 78} However, randomized clinical trials of BACE1 inhibitors (including verubecestat, atabecestat, lanabecestat, and umibecestat) have shown worsening cognitive performance compared with placebo. Some studies have shown significant decline (eg, verubecestat trials: ADAS-Cog worsening v placebo; P<0.05), along with reductions in brain volume on neuroimaging. These findings suggest disruption of normal synaptic function, and consequently, several late stage development programs were discontinued owing to lack of efficacy and safety concerns. Key challenges include achieving selectivity over BACE2, effective penetration of the blood-brain barrier, and favorable pharmacokinetics.⁷⁹ A newer candidate, NP03—a

Table 2 | Phases 2, 3, and 4 clinical trials targeting A β , tau, neuroinflammation, neuromodulation, and metabolic therapies from January 2020-24

NCT number	Drug/treatment	Estimated enrolment (n)	Treatment length (weeks)	Interventions	Primary outcome measure(s)	Trial phase
Anti-amyloid therapy						
NCT05423522	NanoLithium NP03	68	12	NanoLithium NP03 Placebo	NPI-12	2
NCT04498650	Varoglutamstat (PQ912)	259	48 to 96	PQ912 Placebo	Safety	2
NCT04693520	ALZ-801	84	65	ALZ-801	Plasma biomarker of core Alzheimer's disease pathology	2
NCT04770220	ALZ-801	300	78	ALZ-801 Placebo	ADAS-cog 13	3
NCT05531656	CT1812	540	78	CT1812 Placebo	Clinical dementia rating (CDR) sum of boxes (SoB)	2
NCT04735536	CT1812	16	29 days	CT1812 Placebo	Safety EEG Spectral Measures	2
NCT06182085	PRI-002	270	48	PRI-002 Placebo	Safety CDR SoB	2
NCT05181475	Sodium oligomannate (GV-971)	800	48	GV-971	ADAS-cog 11	4
NCT05058040	Sodium oligomannate (GV-971)	2500	96	GV-971	Long term safety	4
NCT05908695	Sodium oligomannate (GV-971)	1312	36	GV-971 Placebo	ADAS-cog 12	4
NCT04520412	Sodium oligomannate (GV-971)	2046	52	GV-971 Placebo	ADAS-cog 11	3
NCT04468659	Lecanemab	1400	216	Lecanemab Placebo	Preclinical Alzheimer Cognitive Composite 5 (PACC5)	3
NCT04639050	Trontinemab (RO7126209)	210	28	RO7126209 Placebo	Safety Amyloid PET	1 2
NCT05026866	Donanemab (LY3002813)	2600	182	LY3002813 Placebo	CDR global score	3
NCT05463731	Remternetug (LY3372993)	600	104	Remternetug (IV) Remternetug (SC) Placebo	Amyloid PET	3
NCT05291234	ABBV-916	195	24	ABBV-916 Placebo	Safety	2
NCT05462106	ACI-24.060	140	48	ACI-24.060 Placebo	Safety	1 2
NCT05834296	ALZN002	30	165	ALZN002 Placebo	Safety	1 2
Tau-targeting therapy						
NCT05399888	Oligonucleotide	735	76	BIIB080 placebo	CDR SoB	2
NCT04524351	Buntanetap (Posiphen)	14	25 days	Posiphen Placebo	Safety	1 2
NCT05422885	Dasatinib+Quercetin	12	12	Dasatinib Quercetin	Neurovascular coupling Executive function	1 2
NCT04685590	Dasatinib+Quercetin	48	48	Dasatinib+Quercetin Placebo	Safety	2
NCT05143528	Nilotinib BE	1275	72	Nilotinib BE Placebo	CDR SoB	3
NCT05194163	MW150	24	12	MW150 Placebo	Safety	2
NCT05063539	LY3372689	330	124	LY3372689 Placebo	Integrated Alzheimer's disease rating scale	2
NCT04445831	ACI-35.030	57	74	ACI-35.030 Placebo	Safety	1 2
NCT04619420	JNJ-63733657	523	104	JNJ-63733657 Placebo	Integrated Alzheimer's disease rating scale	2
NCT04867616	Bepranemab (UCB0107)	421	80	UCB0107 Placebo	CDR SoB	2
NCT04971733	E2814	13	96	E2814	Safety	1 2
NCT05269394	E2814 (+lecanemab open label)	168	208	E2814 Lecanemab Placebo	tau PET	2 3
NCT06268886	BMS-986446	475	76	BMS-986446 Placebo	CDR SoB	2
Targeting multiple neurotoxic proteins						
NCT05686044	Buntanetap	320	12	Buntanetap Posiphen Placebo	ADAS-Cog 11	2 3
Neuroinflammation						
NCT06096090	Interleukin-2	40	26	Interleukin-2 Placebo	Safety	2
NCT05468073	Interleukin-2	45	18	Proleukin Placebo	Change from baseline CDR score	2
NCT04795466	Canakinumab	34	20	Canakinumab Placebo	Neuropsychological test battery	2
NCT05318976	XPro1595	201	23	XPro1595 Placebo	Change in early and mild Alzheimer's cognitive composite	2
NCT05564169	Masitinib	600	24 to 48	Masitinib Placebo	ADAS-cog 11	3
NCT04804241	Senicapoc	55	52	Senicapoc Placebo	ADAS-cog 13	2
NCT04592874	AL002	328	48 to 96	AL002 Placebo	CDR SoB	2
NCT04381468	Pepinemab (semaphorin 4D antibody)	50	44	Pepinemab Placebo	Safety	1 2
NCT05040321	MIB-626	50	90 days	MIB-626 Placebo	CSF concentrations of MIB-626	1 2
Neuromodulation						
NCT04538066	Bryostatin	117	28	Bryostatin 1 Placebo	Safety	2
NCT05986721	AGB101	1040	78	AGB101 Placebo	CDR SoB	3
NCT05771428	ABBV-552	240	12	ABBV-552 Placebo	ADAS-cog 14	2
NCT04838301	Allopregnanolone	200	52	Allopregnanolone Placebo	Hippocampal volume	2
NCT06125951	Xanamem	220	36	Xanamem Placebo	Custom global cognitive test battery	2
NCT04271163	Transcranial electromagnetic treatment	6	52	MemorEM	ADAS-cog 13	1 2
NCT05006781	Transcranial direct current stimulation	140	26	DAOIB	ADAS-cog	2

(Continued)

Table 2 | (Continued)

NCT number	Drug/treatment	Estimated enrolment (n)	Treatment length (weeks)	Interventions	Primary outcome measure(s)	Trial phase
NCT06135051	Deep brain low-intensity ultrasound	40	13	Diadem prototype	Montreal cognitive assessment test for dementia	1 2
NCT04784416	Transcranial photobiomodulation	125	8	Active tPBM-2.0 Sham tPBM-2.0	Repeatable battery for the assessment of neuro-psychological status update	2
NCT04488419	ATH-1017 (fosgonimeton)	554	26	ATH-1017 Placebo	ADAS-cog 11 Alzheimer's disease cooperative study-activities of daily living	2 3
Metabolic						
NCT05006599	Insulin	40	4	Insulin (Humulin R U-100) Placebo	Percentage of prescribed dose taken	2
NCT05081219	Insulin	60	4	Insulin (Humulin R U-100) Empagliflozin Placebo	Safety	2
NCT04777396	Semaglutide	1840	173	Semaglutide Placebo	CDR SoB	3
NCT05891496	Semaglutide	24	77	Semaglutide Placebo	Change in gene expression	3
NCT06072963	Semaglutide	80	52	Semaglutide Intranasal insulin placebo	z-scores of four executive function tests	2
NCT04251182	T3D-959	250	24	T3D-959 Placebo	ADAS-cog 11	2
NCT06052163	Bumetanide	40	26	Bumetanide Placebo	Safety	2
NCT05809908	Tricaprilin	535	26	Tricaprilin Placebo	ADAS-cog	3
NCT06223360	Benfotiamine	406	72	Benfotiamine Placebo	Safety	2

microdose lithium formulation that inhibits GSK-3 β and lowers BACE1 activity—has shown in rodent Alzheimer's disease models to reduce A β load, restore memory, decrease neuroinflammation, and improve synaptic plasticity.^{80 81}

γ -secretase inhibition

γ -secretase inhibitor drugs block presenilin's active site, reducing A β production.⁸² In a phase 3 trial (NCT00594568), semagacestat given daily at doses of 100 mg and 140 mg lowered brain A β but caused adverse effects by also inhibiting Notch signaling, which is essential for developmental and homeostatic functions.⁸³ The trial was terminated early after patients showed worsened cognitive decline despite A β reduction.

γ -secretase modulators

Recent findings have renewed interest in γ -secretase modulator drugs as safer alternatives to inhibitor drugs. These agents selectively lower toxic A β_{42} , while increasing shorter, less amyloidogenic peptides (A β_{38} , A β_{37}), preserving total A β and sparing Notch signaling.⁸² In rodent models, oral modulators reduce and help clear A β plaques.⁸⁴ Early trials of PF-06648671 show good safety and lowered CSF A β_{42} and A β_{40} levels in healthy volunteers.⁸⁵ Although no γ -secretase modulator drugs have advanced to late stage trials, they remain promising for early intervention of Alzheimer's disease.

Targeting A β oligomers

Several agents are in development to inhibit the formation of A β oligomers by stabilizing soluble A $\beta_{40/42}$ monomers,⁸⁶ dismantling toxic oligomers, or reducing their receptor binding to restore synaptic function.^{87 88} ALZ-801, an oral prodrug of tramiprosate and small molecule anti-oligomer

agent, was assessed in two pooled phase 3 trials (2025 participants, 78 weeks).^{89 90} Although the trials did not meet primary endpoints overall, APOE ϵ 4 homozygotes treated with 150 mg twice daily showed cognitive improvement on Alzheimer's disease assessment scale-cognitive subscale (ADAS-cog) scores at weeks 65 and 78 compared with placebo, while heterozygotes and non-carriers showed no benefit.

Targeting A β fibrils

Sodium oligomannate is an oral therapy that binds multiple A β sites to block fibril formation and convert existing fibrils into non-toxic monomers while modulating gut microbiota to reduce peripheral inflammation.⁹¹ Approved in China in 2019 for mild to moderate Alzheimer's disease (450 mg twice daily), it showed mixed outcomes in trials. A 24 week phase 2 study (255 participants), 600/900 mg daily found no overall ADAS-cog 12 improvement, though 900 mg preserved brain glucose metabolism in a subset.⁹² A 36 week phase 3 trial (818 participants), 900 mg noticeably improved ADAS-cog 12 scores versus placebo.⁹³ Ongoing trials aim to support global approval. Table 2 details the phases 2, 3, and 4 clinical trials that have assessed, or are assessing, anti-A β therapies in Alzheimer's disease.

Targeting pyroglutamate A β_{42} production

Pyroglutamate A β_{42} , a toxic derivative of A β formed through glutaminyl cyclase activity, contributes to synaptic damage.⁹⁴ Varoglutamstat, a glutaminyl cyclase inhibitor, reduces this peptide's accumulation and has shown good safety in clinical studies.⁹⁵

Why do certain anti-amyloid strategies work?

Early therapeutic intervention appears crucial for the success of anti-A β therapies, because Alzheimer's

disease pathology begins decades before symptoms arise. Once noticeable neurodegeneration occurs, A β clearance alone might be insufficient to delay cognitive decline. Failures in early trials likely stemmed from enrolling patients at later stages or without confirmed A β pathology. Findings from the Expedition and PRIME trials emphasized the importance of early intervention, A β confirmation, and achieving sufficient clearance thresholds for clinical benefit.^{96 97}

Evidence suggests that both the rate and extent of A β removal determine outcomes, with cognitive benefit emerging when A β levels fall below 40—more reliably under 20—centiloids.^{98 99} Donanemab and lecanemab achieved these levels and slowed progression, while aducanumab and gantenerumab showed variable efficacy linked to inadequate clearance.⁹⁸ Ongoing studies are assessing how antibody binding properties influence both efficacy and side effects.

Although anti-A β therapies modestly delay cognitive decline ($\approx 30\%$), their clinical meaningfulness and durability remain uncertain. Improvements in biomarkers such as CSF p-tau, glial fibrillary acidic protein, and tau-PET indicate downstream effects, though these might reflect indirect responses to A β removal.⁶² Persistent cognitive decline despite A β clearance suggests contributions from tau pathology and other mechanisms, underscoring the need for multi-targeted treatment approaches.⁹⁸

Interpretation of downstream biomarker changes remains complex. For instance, donanemab reduced A β but showed limited tau-PET benefit, while CSF p-tau reductions likely mirror A β load rather than direct tau modulation.¹⁰⁰ Similarly, declines in glial fibrillary acidic protein might reflect reduced astrocytic activation but not necessarily therapeutic recovery.

The approvals of donanemab and lecanemab in 2021-23 were pivotal, they provided the first consistent evidence of modest clinical benefit from effective A β removal.¹⁰¹ However, neither drug halts the progression nor sustains long term stabilization of the disease. Given the multifactorial nature of Alzheimer's disease, combination strategies addressing tau aggregation, neuroinflammation, and synaptic dysfunction are likely essential for achieving meaningful and durable disease modification. The potential health risks of exposing patients to therapies must be carefully weighed.

Tau-targeting therapy

Neurofibrillary tangles, composed of hyperphosphorylated tau, are a central hallmark of Alzheimer's disease and correlate closely with cognitive decline and neuronal loss.^{102 103} Their spread follows a predictable pattern described by Braak staging—beginning in the medial temporal lobe (stage I/II), extending through limbic regions (stage III/IV), and eventually reaching the neocortex (stage V/VI).¹⁰⁴ This progression aligns more strongly with dementia severity than amyloid burden. Beyond

Alzheimer's disease, abnormal tau aggregation defines several tauopathies, including frontotemporal dementia variants, Pick's disease, progressive supranuclear palsy, corticobasal degeneration, and chronic traumatic encephalopathy, each with distinct tau isoform profiles.^{105 106}

The mature human central nervous system expresses six tau isoforms, which arise from the alternative splicing of pre-mRNA transcribed from the microtubule-associated protein tau gene.¹⁰⁷ Cryogenic electron microscopy studies have unveiled near atomic structures of tau filaments, showing disease specific folds that explain the diversity among tauopathies. In Alzheimer's disease, tau containing both 3R and 4R isoforms aggregates into paired helical and straight filaments with a characteristic C shaped protofilament structure. In contrast, Pick's disease, which is composed predominantly of 3R tau, shows a distinct J shaped configuration. Tauopathies involving 4R tau, such as progressive supranuclear palsy and corticobasal degeneration, exhibit alternative filament conformations with different β sheet arrangements. These structural differences likely reflect disease specific molecular contexts, although the exact roles of post-translational modifications and associated cofactors are not yet fully established.¹⁰⁸⁻¹¹⁰ These conformational fingerprints reveal that tau misfolding follows strain-like, templated pathways underlying clinical and pathological heterogeneity.^{111 112} Such insights support the development of isoform selective therapies—such as 4R targeted antibodies or antisense oligonucleotides—and combined 3R/4R strategies for Alzheimer's disease, which advance precision medicine across the tauopathy spectrum.

Under physiological conditions, microtubule-associated protein tau regulates microtubule dynamics and facilitates intracellular transport in neurons.¹¹³ The balance between 3R and 4R tau isoforms is vital for neuronal stability, and disruptions in this ratio are linked to neurodegenerative disorders. Alternative splicing of tau varies across species, with six isoforms expressed in the adult human brain and up to four in rodents.¹¹⁴⁻¹¹⁷ In vitro studies show that 4R tau isoforms aggregate more readily,¹¹⁸ forming oligomers and undergoing post-translational modifications such as phosphorylation, particularly at KxGS motifs, which destabilizes microtubules and damages synapses.¹¹⁹ Studies on astrocytes derived from human induced pluripotent stem cells revealed that both 3R and 4R tau are internalized, though vesicles that contain 3R exhibit greater toxicity.¹¹⁸ In Alzheimer's disease, both isoforms become hyperphosphorylated, forming neurofibrillary tangles that disrupt cytoskeletal and nuclear function.

These findings emphasize the importance of isoform-specific tau-targeting strategies that address aggregation, clearance, and microtubule dysfunction (fig 4). Table 2 lists clinical trials that assess specific tau targeting therapies, with their rationale and mechanism explained below.

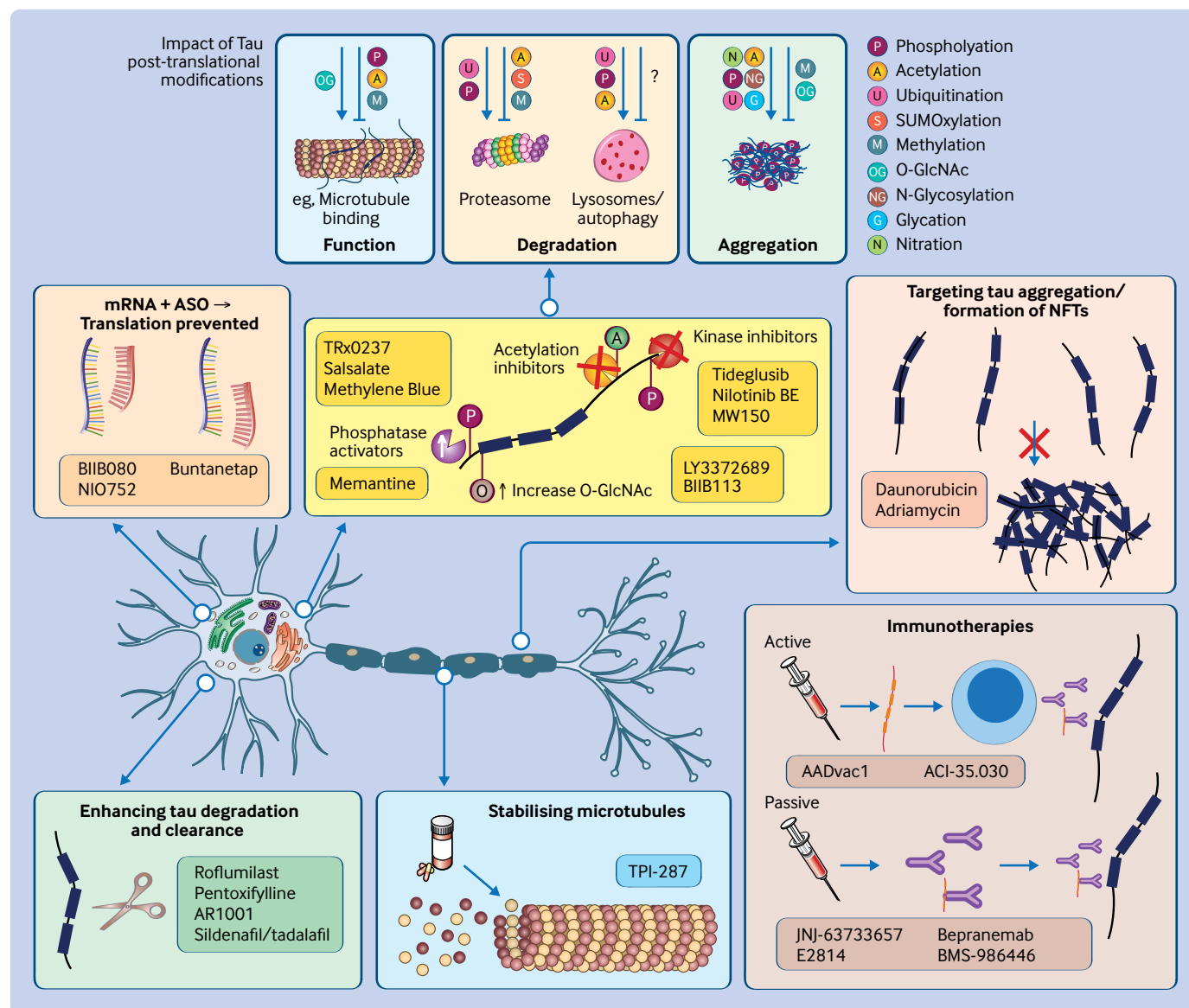


Fig 4 | Tau targeting therapies. Several tau targeting approaches have undergone clinical evaluation including: targeting tau expression (ASOs); targeting post-translational modifications (phosphatase activators, kinase inhibitors, acetylation inhibitors, increasing the O-GlcNAc modification); targeting tau aggregation/formation of neurofibrillary tangles; targeting impairments in tau degradation and clearance pathways; targeting microtubule disruption and dysfunction; and immunotherapies

Targeting post-translational modifications of tau

Tau undergoes multiple post-translational modifications that influence its tendency to form toxic aggregates. In Alzheimer's disease, excessive kinase activity (eg, cyclin dependent kinase 5) and reduced phosphatase function lead to tau hyperphosphorylation, which promotes the formation of fibrils and accumulation of neurofibrillary tangles.¹²⁰ Therapeutic efforts aim to restore balance by inhibiting kinases or enhancing phosphatases. Glycosylation, particularly O-GlcNAcylation, counteracts hyperphosphorylation by competing for similar sites on tau and maintaining its normal function. Increasing O-GlcNAcylation—such as through inhibition of the O-GlcNAcase

enzyme—represents a promising approach to reduce pathological tau accumulation and protect against tau mediated neurodegeneration.¹²¹

Tau targeting immunotherapies

Like A β therapies, tau targeting immunotherapies include both active and passive approaches. Active vaccines stimulate the body's immune system to produce antibodies against abnormal tau. AADvac1, a synthetic tau peptide vaccine, has shown good tolerability and strong IgG responses, with phase 2 analyses suggesting that cognitive decline was slowed in patients that were A β positive and tau positive.^{122–123} Another candidate, ACI-35.030, targets phosphorylated tau epitopes (pSer396, pSer404).

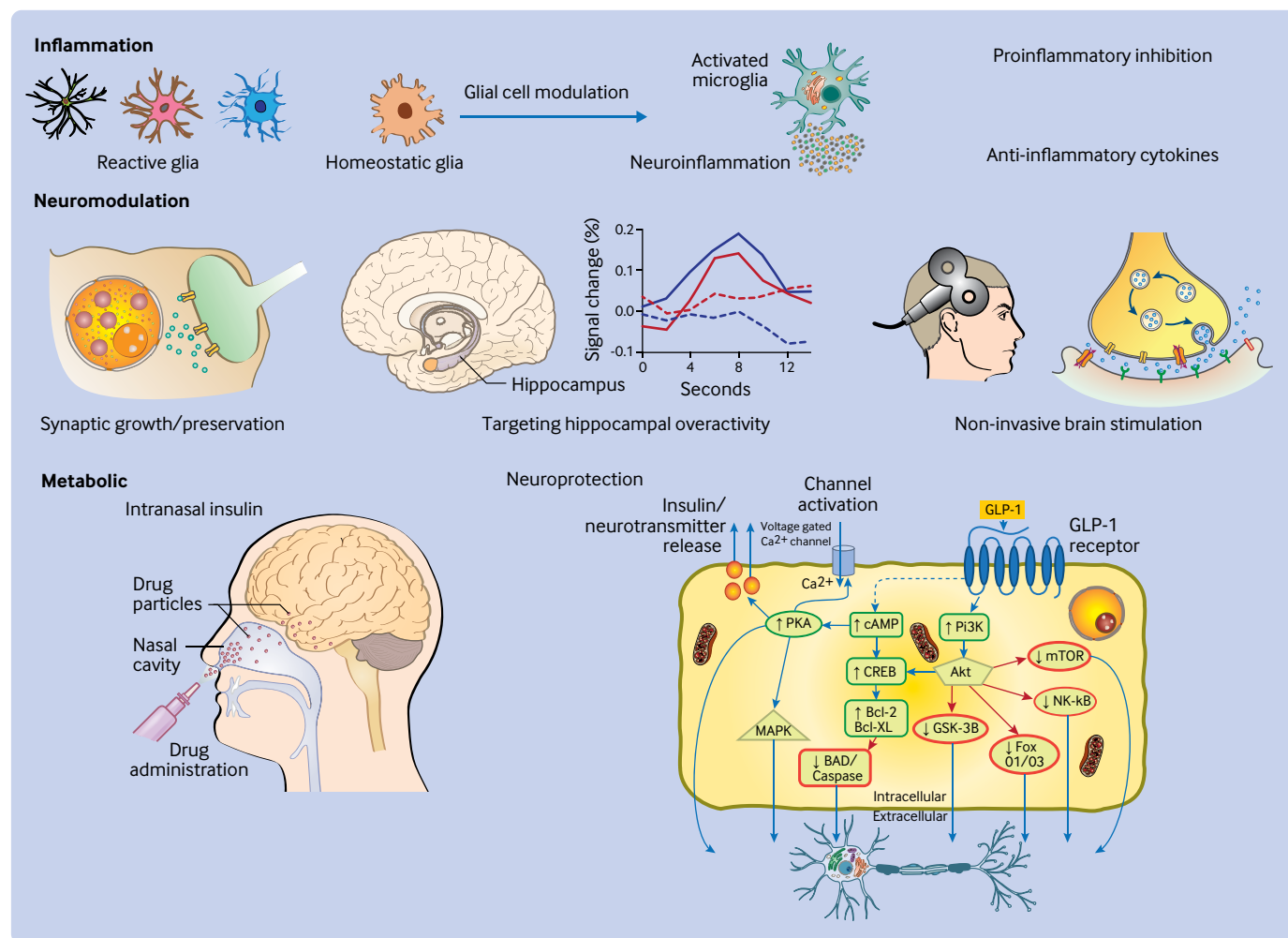


Fig 5 | Additional therapeutic strategies for Alzheimer's disease. Inflammation: targeting neuroinflammation by inhibiting the proinflammatory functions, modulating microglial phenotype to promote anti-inflammatory activity, targeting microglial priming at early stages of disease development. Neuromodulation: therapies include those that promote synaptic growth, preventing hippocampal overactivity, and non-invasive brain stimulation. Metabolic: therapies include intranasal insulin and glucagon-like peptide-1receptor agonists

Passive immunotherapies involve administering antibodies that recognize specific tau regions directly after they have been produced in a laboratory.¹²⁴ Candidates include JNJ-63733657, bepranemab (UCB0107), E2814, and BMS-986446 (PRX005), which bind to distinct tau domains involved in aggregation and spread (table 2).¹²⁵ Several passive anti-tau immunotherapies, including gosuranemab and semorinemab, failed to show clinical efficacy despite reducing extracellular tau levels.^{126 127} These findings suggest that targeting the N terminal region of tau might not sufficiently influence the pathological forms driving neurodegeneration, underscoring the need to explore alternative epitopes and mechanisms in future antibody development. Although several trials have not met primary endpoints, ongoing research emphasizes optimizing antibody characteristics—such as target epitope, isotype, and binding affinity—to enhance therapeutic effectiveness and reduce adverse events.

Targeting neuroinflammation and the role of microglia

The amyloid cascade and tau hypotheses have dominated drug development for Alzheimer's disease, yet modest efficacy, adverse events, and limited accessibility highlight the need for new drugs. Beyond toxic protein accumulation, sustained neuroinflammation is a defining feature of Alzheimer's disease.^{13 128} Microglia and astrocytes drive this process—microglia respond to misfolded A β , while astrocytes support neuronal function but become reactive under chronic inflammation. Persistent glial activation triggers inflammatory cascades involving interleukin 1 β , interleukin 6, interleukin 18, tumor necrosis factor, chemokines, and reactive oxygen species, leading to neuronal loss and synaptic dysfunction.^{13 128} As inflammation is increasingly recognized as central to Alzheimer's disease pathogenesis, about 22% of current investigational drugs target neuroinflammation or

oxidative stress, with several agents now in phases 2-4 clinical trials.¹²⁹ Table 2 lists phases 2, 3, and 4 trials assessing therapies targeting inflammation.

Elucidating the role of microglia in the progression of Alzheimer's disease provides opportunities for therapeutic intervention by using three main approaches: inhibiting the proinflammatory functions of microglia to mitigate the harmful effects of their activation; modulating microglial phenotype to promote anti-inflammatory activity; and targeting microglial priming at early stages of disease development to alter its trajectory (fig 5).

Suppressing proinflammatory properties of microglia

Reducing neuroinflammation in Alzheimer's disease might involve anti-inflammatory compounds or cytokine inhibition. Canakinumab, an antibody that neutralizes interleukin 1 β , is under a 20 week phase 2 trial in participants with mild cognitive impairment or mild Alzheimer's disease with peripheral inflammation (NCT04795466). Low dose interleukin 2 (1.5 million IU administered subcutaneously, three times daily for five consecutive days every four weeks) showed anti-inflammatory and neuroprotective effects in rodents and was well tolerated in a phase 2a trial, expanding regulatory T cells, lowering proinflammatory mediators, and improving CSF A β_{42} , with cognitive trends favoring slower decline though not statistically significant (P=0.061).^{130 131}

NE3107 is an orally active small molecule that modulates insulin signaling and suppresses neuroinflammation by inhibiting extracellular signal-regulated kinase and nuclear factor kappa-light-chain-enhancer of activated B cell mediated pathways. It aims to reduce inflammatory insulin resistance and improve neuronal metabolism. Phase 3 trials are ongoing to evaluate its efficacy in slowing cognitive decline and disease progression in Alzheimer's disease.¹³²

Masitinib, a selective tyrosine kinase inhibitor drug that reduces mast cell mediated cytokine release, improved cognition in Alzheimer's disease.¹³³ In a phase 2 trial (34 participants), oral masitinib given twice daily for 24 weeks improved ADAS-cog scores versus placebo. In a phase 3 study (358 participants), masitinib 4.5 mg/kg/day again showed benefit over placebo.¹³⁴

Earlier trials with non-steroidal anti-inflammatory drugs were ineffective, likely owing to their broad, non-specific suppression of both proinflammatory and anti-inflammatory responses.¹³⁵ These results underscore the need for targeted therapies that selectively modulate detrimental inflammatory pathways in Alzheimer's disease.

Modulating microglial phenotypic change in favor of the anti-inflammatory properties

In early Alzheimer's disease, microglia help maintain neuronal health by clearing toxic material, but sustained activation drives inflammation. Modulating microglial function to promote anti-inflammatory,

phagocytic activity offers therapeutic promise. The KCa3.1 channel inhibitor senicapoc mitigates Alzheimer's disease-like deficits in the five mutations associated with familial Alzheimer's disease mouse model,¹³⁶ while AL002, a TREM2 monoclonal antibody, enhances microglial proliferation and has shown good tolerability in humans.¹³⁷ Targeting glial responses also includes pepinemab, a monoclonal antibody against semaphorin-4D that limits reactive astrogliosis and showed a favorable safety profile in a phase 2 trial in Huntington's disease,^{138 139} which supports its potential relevance to Alzheimer's disease.

Influencing microglial priming early in the disease trajectory

Microglial priming, associated with aging and systemic inflammation, leads to exaggerated immune responses in the brain.¹⁴⁰ Preventing this priming during the preclinical phase of Alzheimer's disease could mitigate neuroinflammation. Raised levels of low density lipoprotein in Alzheimer's disease might also promote vascular inflammation by endothelial activation.¹⁴¹ MIB-626, an oral β -nicotinamide mononucleotide polymorph, improves glucose and lipid metabolism and mitochondrial function, showing good tolerability in a placebo controlled trial of 32 participants who were overweight.¹⁴² Dietary interventions, including folic acid and omega 3 fatty acid supplementation, have further shown potential to reduce inflammatory biomarkers in Alzheimer's disease.^{143 144}

Neuromodulation

Synaptic and neuronal dysfunction are central to Alzheimer's disease, with accelerated neuronal loss beginning even in preclinical stages.¹⁴⁵ Multiple therapeutic strategies are being developed to ameliorate synaptic and neuronal function because synaptic loss strongly correlates with cognitive decline (fig 5).¹⁴⁶

Blarcamesine is an orally available sigma-1 receptor agonist that aims to restore cellular homeostasis by using pathways including autophagy enhancement and mitochondrial stabilisation. In a recent 48 week phase 2b/3 trial in early Alzheimer's disease, blarcamesine substantially slowed cognitive decline and brain atrophy relative to placebo, with a favorable safety profile and no adverse effects related to imaging.¹⁴⁷

Bryostatin is a natural product that activates protein kinase C to enhance synaptic numbers. Results of bryostatin efficacy have received mixed results in two phase 2 trials.^{148 149} A pooled analysis of 94 participants with moderate to advanced Alzheimer's disease showed that bryostatin, in the absence of memantine, improved the severe impairment battery score by week 13.¹⁵⁰ A phase 2 trial showed benefit of bryostatin versus placebo in individuals with a moderate-severe mini mental state examination score (10-14) on the severe impairment battery score at week 28.¹⁵¹

AGB101 is an inhibitor of the SV2A protein that targets hippocampal overactivity in early Alzheimer's disease. Phase 2 investigation showed no benefit of AGB101 compared with placebo on clinical dementia rating sum of boxes scores.¹⁵² In a pre-specified analysis, APOE ϵ 4 non-carriers showed a relative improvement of 40%, compared with 10% in APOE ϵ 4 carriers, however, the difference was considered not substantial.¹⁵²

Non-invasive brain stimulation

Non-invasive brain stimulation is a rapidly emerging technique that inhibits or excites neuronal activity to exert a therapeutic benefit.¹⁵³ Transcranial electromagnetic stimulation generates an electromagnetic field that excites neuronal cells affected by the magnetic field. A short lasting electrical current is passed through the loops of conducting wire, which is delivered to the brain to depolarize neurons.¹⁵⁴

A total of eight participants with mild to moderate Alzheimer's disease completed a clinical trial of two months of daily transcranial electromagnetic treatment.¹⁵⁵ Transcranial electromagnetic treatment revealed a trend toward improved ADAS-cog 13 performance compared with baseline at two months. After the removal of one non-responder, the treatment noticeably improved ADAS-cog 13 at two months and 14 days post-treatment in the remaining seven participants.

Another approach involves transcranial magnetic stimulation that uses magnetic fields to induce electrical currents in targeted brain regions, thereby modulating neuronal activity. A recent phase 2, double blind, sham controlled trial showed that 24 weeks of precuneus repetitive transcranial magnetic stimulation substantially slowed cognitive decline in patients with mild to moderate Alzheimer's disease.¹⁵⁶

Transcranial direct current stimulation provides a cheap, portable alternative that delivers low intensity current.¹⁵⁷ Meta-analysis of 14 case-control/clinical studies evaluated the influence of transcranial direct current stimulation in participants with Alzheimer's disease. Positive effects were shown on general cognition and memory, although the influence on cognitive subdomains showed high heterogeneity between studies.¹⁵³

Table 2 details the phases 2, 3, and 4 clinical trials that have assessed therapies that target neuromodulation in Alzheimer's disease.

Neuromodulation therapy for Alzheimer's disease currently faces several challenges, including limited long term efficacy data, variable patient responses, and uncertainty about optimal stimulation parameters and targets. Future research should focus on refining patient selection by using neurophysiological and biomarker-based profiling, integrating neuromodulation with pharmacological or cognitive interventions, and establishing standardized protocols to enhance reproducibility and clinical benefit. Advancing these areas will be essential to move neuromodulation from an experimental application toward reliable therapeutic practice.

Metabolic therapies

Deficient cerebral glucose metabolism is an established hallmark of Alzheimer's disease, with patients also showing deficiencies in peripheral and central insulin signaling.¹⁵⁸ The brain utilizes about 20% of the body's total glucose supply, however, during the early stages of Alzheimer's disease glucose utilization can be reduced by 45%.¹⁵⁹ Deficits in hippocampal glucose metabolism are apparent before the appearance of clinical symptoms.¹⁵ Metabolic decline within the entorhinal cortex can predict conversion from cognitively normal to mild cognitive impairment.¹⁶⁰ At a cellular level, glucose transporter deficiencies that occur in the brain of patients with Alzheimer's disease accentuate disease pathology.¹⁵ Alzheimer's disease has been suggested by some as a metabolic disease related to age and coined as type 3 diabetes.¹⁶¹ Strategies based on targeting insulin signaling dysfunction, use of diabetes drugs and dietary interventions demonstrate promise in the search for a novel therapeutic strategy for Alzheimer's disease (fig 5).¹⁶²

Insulin receptors are widespread in the brain and activate key signaling cascades including phosphatidylinositol 3 kinase, AKT, and mechanistic target of rapamycin pathways with crucial roles in memory processing, cell metabolism, neuronal survival, synaptic plasticity, and neuroprotection.¹⁶³ Postmortem analysis of Alzheimer's disease brain tissue has revealed impaired insulin signaling, marked by raised serine phosphorylation of insulin receptor substrate 1 in the hippocampus and cortex, even in the absence of peripheral insulin resistance.¹⁶⁴ This abnormal insulin receptor substrate 1 phosphorylation is linked to tau hyperphosphorylation and A β plaque formation.¹⁶⁵ In patients with Alzheimer's disease, neuronal derived exosomes reveal abnormal insulin signaling, including increased serine and reduced tyrosine phosphorylation of insulin receptor substrate 1, detectable up to 10 years before clinical onset.¹⁶⁶

Insulin

In a pilot trial enrolling 104 participants with amnesic mild cognitive impairment or mild to moderate Alzheimer's disease, 20 IU intranasal insulin improved delayed memory compared with placebo.¹⁶⁷ The co-primary analysis showed that 20 IU and 40 IU insulin improved dementia severity rating scores. A larger trial enrolling 260 participants with amnesic mild cognitive impairment or Alzheimer's disease evaluated the influence of 40 IU intranasal insulin for a period of 12 months.¹⁶⁸ No substantial differences were observed in the primary outcome of the ADAS-cog 12 score between the insulin and placebo treatment groups.

GLP-1 receptor agonists

Glucagon-like peptide-1 (GLP-1) receptor agonists are an approved therapeutic strategy for diabetes and obesity, which activates incretin signaling pathways that effectively modulate blood glucose levels and

importantly do not induce insulin desensitization in individuals who are normoglycemic.¹⁶⁹ GLP-1 receptor agonists operate by using a plethora of mechanisms including attenuating inflammation, reducing tau phosphorylation, and ameliorating synaptic function, in addition to their effect on insulin resistance.¹⁶⁹

A 26 week pilot trial in 38 participants with Alzheimer's disease showed that the GLP-1 receptor agonist, liraglutide, provided a nonsignificant increase in cerebral glucose metabolism ($P \geq 0.49$).¹⁷⁰ No differences were shown in the rate of A β deposition in comparison with placebo. The trial Evaluating Liraglutide in Alzheimer's disease (ELAD) completed a 12 month phase 2b clinical trial of liraglutide in 204 patients with Alzheimer's disease.^{171 172} The primary endpoint for the study, which was change in the cerebral glucose metabolic rate, was not met (difference -0.17 ; $P=0.14$). However, the secondary endpoint of change in scores for cognitive function measured by ADAS-Exec measures ($P=0.01$), as well as the exploratory measure of brain volumes were found to be significant ($P=0.002$). Liraglutide was found to reduce shrinkage of brain volume by 50% compared with placebo ($P<0.016$).¹⁷¹

Similar to liraglutide, semaglutide is an acylated GLP-1 analog that can reverse A β -induced suppression of autophagy, as shown by in vitro studies,¹⁷³ and increase expression of sirtuin 1, which is a protein implicated in multiple cellular processes including glucose metabolism, energy regulation, cellular stress response, and GLUT4 expression. Semaglutide has been shown to mitigate A β and tau pathology, accompanied with ameliorated learning and memory in mouse models of Alzheimer's disease. Semaglutide decreased systemic inflammation in patients and exhibited anti-inflammatory effects by inhibiting atherosclerotic plaques in mouse models devoid of APOE and low density lipoprotein receptor.¹⁷⁴⁻¹⁷⁶ Two phase 3 trials are investigating oral semaglutide, a GLP-1 receptor agonist, in Alzheimer's disease (NCT04777396, NCT05891496). Participants were randomly assigned (1:1) to once daily semaglutide 14 mg (flexible dose) or placebo for up to 156 weeks. The primary endpoint was change in clinical dementia rating sum of boxes score from baseline to week 104, assessed in all randomized participants. 3808 were randomly assigned; 1855 in evoke (semaglutide, $n=928$; placebo, $n=927$) and 1953 in evoke+ (semaglutide, $n=976$; placebo, $n=977$). Mean age was 72.2 years (SD 7.1) and mean clinical dementia rating sum of boxes score was 3.7 (SD 1.6) at baseline. In evoke+, 54 (2.8%) participants had small vessel pathology. In evoke and evoke+, mean changes in clinical dementia rating sum of boxes score from baseline to week 104 were 2.3 (SE 0.1) and 2.2 (0.1) with semaglutide, compared with 2.3 (0.1) and 2.1 (0.1) with placebo (estimated difference -0.08 , 95% confidence interval -0.35 to 0.20 , $P=0.57$ in evoke and 0.10 , -0.17 to 0.38 , $P=0.46$ in evoke+). Hence, oral semaglutide was not efficacious in slowing clinical progression in participants with

early Alzheimer's disease.¹⁷⁷ A phase 2 trial is investigating combined semaglutide and insulin therapy in 80 older adults with metabolic syndrome or mild cognitive impairment (NCT06072963).

Triglycerides

Ketone bodies can provide an alternate source of energy for the brain in glucose deficiency.¹⁷⁸ Tricaprilin is the oral formulation of caprylic triglyceride, a source of ketone bodies to improve mitochondrial metabolism. Meta-analysis revealed a benefit of medium chain triglycerides compared with placebo in participants with mild cognitive impairment or Alzheimer's disease, with a greater effect in APOE $\epsilon 4$ non-carriers.¹⁷⁸ However, a clinical trial of tricaprilin showed no benefit on ADAS-cog 11 scores relative to placebo.¹⁷⁹

Benfotiamine

Benfotiamine is an orally available synthetic thiamine precursor that influences multiple metabolic enzymes that are critical for cerebral glucose metabolism.¹⁸⁰ A phase 2 trial investigated the utility of benfotiamine treatment.¹⁸¹ Those treated with benfotiamine showed a lower increase in ADAS-cog 11 score of 43% compared with placebo, although the difference was not statistically significant ($P=0.125$). Benfotiamine was indicated to improve clinical dementia rating scores relative to placebo ($P=0.034$). Table 2 outlines clinical trials examining metabolic drug targets in Alzheimer's disease.

Viral infections

Emerging evidence suggests that viral infections might contribute to the pathogenesis of Alzheimer's disease. Herpesviruses found in the brains of patients with Alzheimer's disease can induce inflammation, oxidative stress, and A β accumulation, supporting the "antimicrobial protection hypothesis" that A β traps pathogens but promotes plaque formation.^{182 183} Other viruses, including hepatitis viruses and severe acute respiratory syndrome coronavirus-2, might also influence Alzheimer's disease processes, with coinfections worsening neurodegeneration. Although causality remains unproven, these links highlight potential for antiviral or vaccine based strategies. Supporting this, a large Welsh study found herpes zoster vaccination reduced dementia risk by 20% over seven years ($P=0.019$), suggesting a feasible preventive approach.¹⁸⁴

Non-drug treatment

The Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability (FINGER) was a two year randomized controlled study involving 1260 older adults at increased risk of cognitive decline. Participants were assigned to a multidomain intervention—including dietary guidance, physical exercise, cognitive training, and management of vascular risk factors—and showed greater improvement in overall cognitive performance

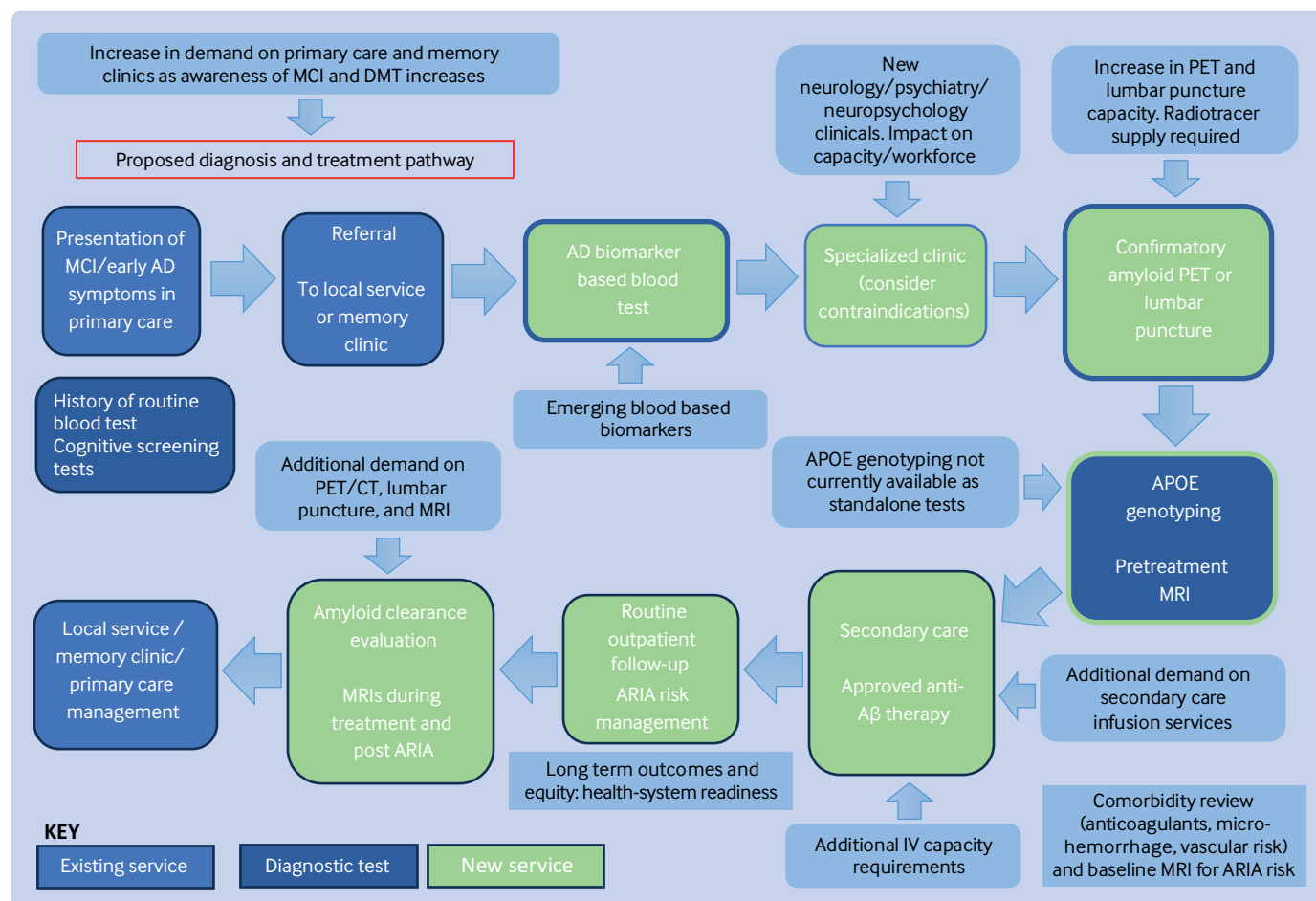


Fig 6 | Real world patient pathway for anti-Aβ therapies. Proposed real world patient pathway for anti-Aβ therapies, outlining diagnostic evaluation, biomarker use, patient selection, ARIA monitoring, and treatment decision points. The pathway spans early clinical suspicion through biomarker confirmation, eligibility assessment (including APOE genotyping and MRI safety checks), treatment initiation within specialist centers, and ongoing cognitive, biomarker, and MRI monitoring. It highlights multidisciplinary collaboration across primary care, memory clinics, and neurology services to enable safe, effective, and equitable delivery of disease modifying therapies. Each stage of the diagnostic pathway serves as a decision point guided by clinical judgment and patient preference. Blood based biomarkers help triage who should undergo confirmatory PET or CSF testing, while others continue routine follow-up, reflecting real world variation where not all patients progress through every step

compared with those receiving standard health advice (composite cognitive score difference ≈ 0.13 ; $P=0.03$). Benefits were most evident in domains such as executive function ($P=0.04$) and processing speed ($P=0.03$).^{185 186}

A separate 20 week multidomain lifestyle intervention in individuals with mild cognitive impairment or early Alzheimer's disease (approximately 50 participants) reported stabilization or modest improvement in cognitive outcomes relative to baseline. Although there was a numerical improvement in measures such as ADAS-Cog (approximately -1.5 points), these changes were not statistically significant ($P=0.053$).^{185 186}

Proposed patient treatment pathway

Clinical expertise remains central to the diagnosis of Alzheimer's disease, with biomarker testing conducted under clinician supervision to support evaluation of cognitive impairment. Rapid progress

in fluid based biomarkers enhances diagnostic potential, but clinical judgment is critical when results conflict with presentation or are influenced by confounders such as trauma, medications, kidney dysfunction, or CSF disorders.^{17 187-194} Biomarkers mainly aid trial participant selection and are not recommended for cognitively normal individuals outside research settings. Standardization of biofluid assays, tau PET methods, and biomarker cut-offs is needed, alongside real world studies to validate findings. As biologically based diagnosis transitions into clinical use, accessible blood based biomarkers will be vital, particularly in low and middle income countries, to enable equitable early detection and guide emerging non-Aβ targeted therapies.

Implementing disease modifying immunotherapies in Alzheimer's disease requires a clear, evidence based clinical pathway that links early recognition, accurate diagnosis, and safe long term management (fig 6). Initial identification begins in

primary care, where cognitive screening tools (eg, Montreal cognitive assessment, mini mental state examination) and exclusion of reversible causes help flag potential candidates for referral to memory clinics. A thorough clinical diagnosis, supported by detailed neuropsychological assessment, remains vital to distinguish Alzheimer's disease from other dementias and to define cognitive stage and functional impact.

Confirmatory biomarkers—PET or CSF assays, and increasingly plasma A β and phosphorylated tau tests—verify amyloid pathology and guide patient selection, supported by standardized laboratory and imaging processes. Eligibility assessment should include comorbidity review, baseline MRI for ARIA risk, and APOE genotyping to personalize counselling and risk-benefit discussions. Patients taking anticoagulant drugs or with extensive microhemorrhage require additional multidisciplinary review.

Treatment must be provided in specialist centers with access to MRI and trained staff for ARIA detection and management, using scheduled scans at 3, 6, and 12 months and clear communication between neurologists, radiologists, and primary care. Decisions on continuation or discontinuation should follow regular 6-12 month clinical and biomarker reassessments, considering stability, adverse events, or evolving comorbidities. Scalable delivery also depends on investment in infusion services, imaging capacity, skilled personnel, and integrated referral

pathways to ensure equitable, safe adoption of anti-A β therapy in routine practice.

From a public health perspective, anti-A β therapies produce modest short term slowing of cognitive decline but are associated with frequent adverse events, uncertain long term benefits, and substantial treatment burden for patients and caregivers. Shared decision making is crucial, with clear discussion of potential risks, treatment logistics, and the need to modify existing drugs—such as discontinuing anticoagulant drugs to reduce bleeding risk. Participants enrolled in clinical trials were relatively young and had early stage Alzheimer's disease with few comorbidities, contrasting sharply with the complex, multimorbid populations seen in practice. Real world data show high rates of mixed dementia, depression, and vascular disease, which could diminish treatment efficacy and increase the risk of adverse events such as ARIA.¹⁹⁵ Consequently, eligibility for current anti-A β therapies remains limited—only about 8% of patients with mild cognitive impairment or mild dementia meet trial criteria—and implementation will demand substantial healthcare resources and costs for a small subset of individuals.⁵¹

Guidelines

Guidelines for Alzheimer's disease share common foundations (such as structured clinical assessment, exclusion of reversible causes, and the use of

Table 3 | Key recommendations from international Alzheimer's disease and mild cognitive impairment guidelines (updated with biomarker based frameworks)¹⁹⁸

Recommendation	Guidelines	Rationale
Screening		
Assess cognitive concerns as possible mild cognitive impairment rather than normal ageing	American Academy of Neurology (AAN)	Mild cognitive impairment reflects underlying pathology and increased dementia risk
Use brief cognitive screening tools (eg, mini mental state examination score)	Korean	Provides acceptable diagnostic accuracy as part of assessment
Consider targeted screening in high risk groups	Traditional Chinese Medicine	Based on limited evidence and expert consensus
Avoid screening asymptomatic individuals	Ministry of Health Singapore Clinical Practice Guidelines (MOHM)	Unclear balance between benefits and harms
Diagnosis		
Exclude reversible or alternative causes	AAN; NICE; Danish; Indian	Reduces misdiagnosis and identifies treatable conditions
Interpret cognitive testing within clinical context	Multiple	Results influenced by demographic and clinical factors
Use biomarker based criteria (positron emission tomography, cerebrospinal fluid, plasma)	Alzheimer's Association	Shift toward pathology based diagnosis
Use blood based biomarkers in specialist settings	Alzheimer's Association	Improves diagnostic access and accuracy
Reserve biomarkers for selected cases	AAN; NICE; Danish	Avoid overdiagnosis
Consider cerebrospinal fluid/amyloid imaging if uncertainty	NICE; Danish; Korean	Improves diagnostic confidence
Assess activities of daily living	Danish; Indian	Supports care planning
Avoid routine APOE testing	NICE; Korean	Risk marker, not diagnostic
Use imaging when indicated	NICE; Danish; Korean; European Association of Nuclear Medicine (EANM)	Improves diagnostic accuracy
Treatment		
Use cholinesterase inhibitors $\pm$ memantine	NICE; Taiwan; Indian; Australian	Modest benefit
Do not use pharmacotherapy for mild cognitive impairment	AAN; Taiwan; Indian	No proven benefit
Recommend exercise and cognitive interventions	AAN; WHO; Taiwan	Might improve cognition
Use non-drug approaches first	Multiple	Safer, improves quality of life
Monitoring		
Monitor cognition over time	AAN	Variable progression
Review cognition, drugs, comorbidities	Irish	Supports care
Follow subjective cognitive decline	Korean	Higher risk progression
Do not stop cholinesterase inhibitors based on severity	NICE	No stopping threshold
If deprescribing, monitor closely	Australian	Minimizes harm

cognitive testing), but differ in how they approach diagnosis and emerging technologies. Earlier guidance, including that from the American Academy of Neurology and the National Institute for Health and Care Excellence, primarily follows a clinical model, where diagnosis is based on symptoms, and biomarkers are reserved for selected cases with diagnostic uncertainty. In contrast, more recent recommendations from the Alzheimer's Association reflect a shift toward a biologically defined framework, in which Alzheimer's disease can be identified by using abnormal biomarker findings even at earlier stages.¹⁹⁶⁻¹⁹⁹

Differences are particularly evident in the use of biomarkers, especially blood based tests. Traditional guidelines either limit or do not include these tools, whereas newer recommendations support their use in specialist settings as part of the diagnostic pathway, provided they meet defined performance standards. Guidance also varies regarding genetic testing and screening strategies, with some recommending selective use in higher risk individuals and others discouraging routine application.¹⁹⁶⁻¹⁹⁹

Despite these variations, there is general agreement across guidelines that currently available drug treatments are symptomatic rather than disease-modifying, that non-pharmacological strategies are essential, and that ongoing monitoring is important. Overall, the main distinction lies in the transition from a symptom based approach to one increasingly supported by biomarkers, with differences reflecting how quickly this shift is being adopted in different regions (table 3).¹⁹⁶⁻¹⁹⁹

Recent Alzheimer's disease guidelines increasingly emphasize early and preclinical diagnosis, driven by advances in biological and clinical assessments.¹⁷ This shift follows repeated failures of trials that enrolled patients at later stages of the disease. International working groups now recommend biomarker based diagnosis, focusing on individuals showing clear Alzheimer's disease related pathophysiology through validated biomarkers and characteristic phenotypes. However, PET and CSF testing remain limited by high cost, invasiveness, and standardization challenges, restricting large scale use. Therefore, optimizing a global framework for biomarker guided, targeted interventions is essential. Emerging blood based biomarkers reflecting core Alzheimer's disease pathology could serve as practical triage tools to prioritize individuals for confirmatory PET imaging.¹⁸⁷

The Alzheimer's Association Workgroup research framework describes a six stage numerical clinical staging scheme for individuals who are in the Alzheimer's disease pathophysiological continuum.¹⁷ The six clinically defined stages are: (1) biomarker evidence of Alzheimer's disease in asymptomatic individuals; (2) transitional decline, which denotes the earliest detectable clinical symptoms possibly owing to Alzheimer's disease in individuals who are cognitively unimpaired; (3) objective cognitive impairment of insufficient severity

QUESTIONS FOR FUTURE RESEARCH

- What is the optimal stage to target the pathological hallmarks of Alzheimer's disease?
- Why do only certain anti-amyloid targets work?
- How can residual cognitive decline that occurs after anti-amyloid immunotherapy be prevented?
- How can the multifaceted nature of Alzheimer's disease be effectively targeted?

resulting in noticeable functional loss (ie, inefficient activities of daily living but still independent); and (4-6) loss of independence with progressively worse functional loss. Alzheimer's disease is defined as a biological process beginning with the appearance of Alzheimer's disease neuropathological change despite individuals remaining asymptomatic. As the neuropathological burden progresses, this results in occurrence and worsening of clinical symptoms.^{17 200} Early changing core biomarkers are A β -PET, approved CSF biomarkers, and accurate plasma biomarkers such as p-tau 217. Abnormalities in the early core biomarkers are sufficient to establish an Alzheimer's disease diagnosis and inform clinical decision across the Alzheimer's disease continuum. Later changing core biomarkers such as biofluid and tau PET are informative in deciphering prognosis, and if found to be abnormal, increase the confidence in Alzheimer's disease driving the disease process and its clinical manifestation.

Conclusion

Emerging Alzheimer's disease drugs aim to relieve symptoms and slow disease progression, with anti-A β treatments offering cautious optimism. Effective management will likely require combination approaches targeting tau, inflammation, neuronal, synaptic, and metabolic dysfunctions, currently in late stage development. Future research should prioritize the validation of blood based biomarkers across heterogeneous, real world populations to ensure clinical reliability and equitable access. Long term outcome studies are needed to determine the durability of cognitive benefit and safety of disease modifying therapies, particularly regarding ARIA management. Parallel efforts should advance tau targeted and combination approaches to address non-amyloid pathology. Research on implementation that should examine health system capacity, workforce training, and cost effectiveness will be crucial to translate therapeutic advances into sustainable clinical practice. Together, these directions will shape a more comprehensive, personalized, and scalable framework for the treatment of Alzheimer's disease.

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